LEARNING WHO TALKS TO WHOM: ROLE-AWARE COMMUNICATION TOPOLOGY SCHEDULING FOR EFFICIENT LLM MULTI-AGENT COLLABORATION

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ABSTRACT

Multi-agent systems driven by large language models (LLMs) rely on structured communication topologies to facilitate information exchange and collaborative reasoning. However, effectively scheduling these topologies to dynamically determine *who talks to whom* and *when to communicate* remains a fundamental challenge in balancing collective reasoning performance against exponential token overhead. Existing topology scheduling approaches either enforce static communication graphs that cannot adapt to dynamic problem-solving states, or overlook agent role complementarity while failing to explicitly incorporate the synthesized graph structure into team-level decision evaluation. To tackle these limitations, we propose `ACTS`, a role-aware adaptive communication topology scheduling framework for efficient LLM multi-agent collaboration. Orchestrated within a Centralized Training with Decentralized Execution (CTDE) paradigm, `ACTS` introduces a novel dual-attention mechanism. Specifically, a *Role-conditioned Relational Attention* module dynamically selects variable-cardinality interaction partners based on agent roles and evolving execution states, while a *Topological Attention Mixer* explicitly embeds the directed communication topology into team value decomposition for precise credit assignment. Comprehensive evaluations across 9 challenging benchmarks covering code generation, multi-step reasoning, general knowledge, and competitive mathematics demonstrate that `ACTS` consistently outperforms state-of-the-art baselines, boosting average accuracy by up to 6.19 percentage points while reducing token consumption by over 15%.

1 INTRODUCTION

LLM-based multi-agent systems coordinate agents with heterogeneous roles and complementary reasoning perspectives to solve complex tasks (Du et al., 2023b; Wu et al., 2023a; 2026). This specialization can expand the evidence available to the team and support independent cross-agent verification, but it also requires repeated message exchange, thereby increasing token consumption and inference latency (Zhu et al., 2026). Communication that is not aligned with the task state or agent roles may generate redundant messages without improving the recipient's decision context. These observations motivate the study of role-aware adaptive communication topology scheduling for efficient LLM-based collaboration. The scheduler must jointly determine when communication occurs and which directed agent pairs exchange messages, conditioned on the evolving task and interaction states while controlling token overhead.

Existing methods address communication topology at different levels. Fixed workflows and predefined communication patterns apply similar interaction structures across tasks and rounds (Du et al., 2023b; Wu et al., 2023a), which may generate redundant exchanges. Task-adaptive topology-generation and architecture-search methods condition the graph on the task, but typically select it before execution and do not update communication links as local interaction states evolve (Zhang et al., 2024a; Sun et al., 2026; Jiang et al., 2025b; Zhang et al., 2025a). Agent Q-Mix introduces round-wise communication-mode decisions and combines per-agent utilities with a QMIX-based value-decomposition mixer under CTDE (Jiang et al., 2026a; Rashid et al., 2018a). However, it does not explicitly incorporate agent-role information into communication scheduling or value factorization, and its mixer is not conditioned on the realized directed communication topology. Con-

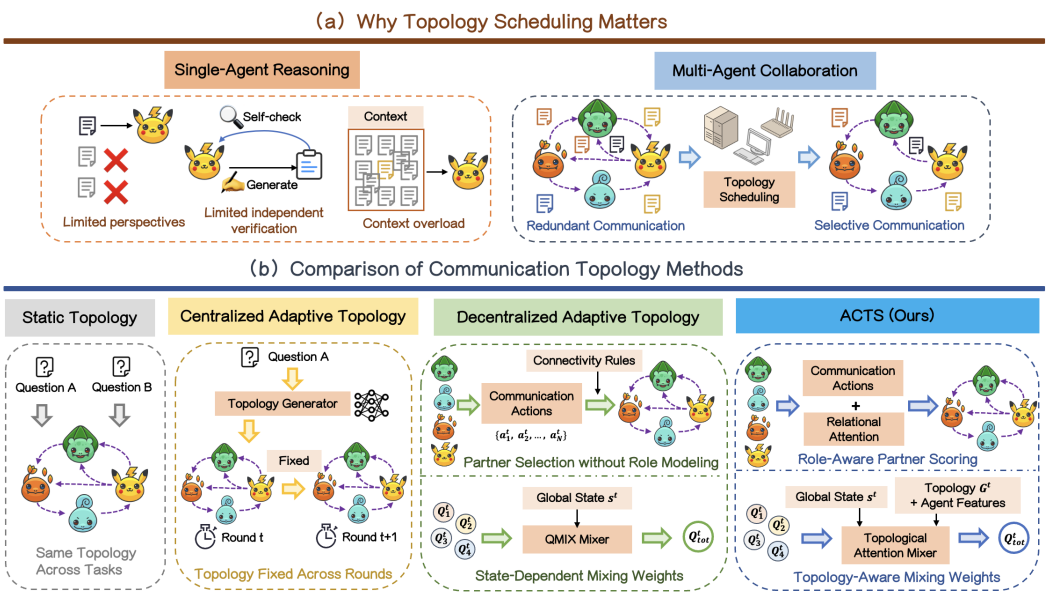


Figure 1: **Motivation and method comparison.** (a) Multi-agent collaboration addresses limitations of single-agent reasoning, while topology scheduling improves communication efficiency by regulating message exchanges. (b) Representative methods differ in topology adaptation, partner selection, and value-function factorization. ACTS uses relational attention for role-aware partner selection and topological attention to learn topology-aware mixing weights.

sequently, the joint-value mixer does not explicitly condition an agent's contribution on its role or on the directed incoming and outgoing neighborhoods induced by communication. This architectural distinction is illustrated in Figure 1(b).

We introduce ACTS, a role-aware framework for **A**daptive **C**ommunication-**T**opology **S**cheduling in LLM-based multi-agent collaboration. ACTS separates communication-mode selection from partner selection. At each round, local action-value functions select the communication mode, while role-conditioned relational attention assigns directional relevance scores to candidate partners using role embeddings and recurrent local states. The selected modes and partner sets define a variable-cardinality directed topology. Under CTDE, a topological attention mixer encodes the directed incoming and outgoing neighborhoods of the pre-decision graph into joint-value function factorization, thereby learning topology-dependent utility weights for centralized credit assignment. Across nine benchmarks, ACTS achieves the highest macro-average accuracy among the evaluated methods on both backbone LLMs, reaching 75.01% with GPT-OSS-120B and 69.43% with DeepSeek-V4-Flash. Relative to Agent Q-Mix, it improves average accuracy by 3.18 and 6.19 percentage points while reducing average token consumption by 15.42% and 13.84%, respectively. Ablation studies further show that removing role embeddings or learned attention reduces accuracy and increases token consumption.

To summarize, our contributions are as follows:

- We formulate communication-topology scheduling for LLM-based multi-agent systems as a role-aware sequential decision problem that jointly selects communication modes and directed partner sets from agent roles and evolving local states.
- We develop ACTS, a CTDE-based value-decomposition framework that combines role-conditioned relational attention for directed partner selection with a topological attention mixer that conditions local-utility weights on the realized pre-decision topology, enabling topology-aware centralized credit assignment.
- We evaluate ACTS on nine benchmarks with two frozen LLM backbones. ACTS achieves the highest macro-average accuracy among the evaluated methods, improving over Agent Q-Mix by up to 6.19 percentage points while reducing token consumption by up to 15.42%. Ablation studies further validate the contributions of role-aware representations and learned attention.

2 Related Work

2.1 Communication and Topology Control

LLM-based multi-agent systems initially relied on fixed role-based workflows, structured debates, and predefined interaction templates, including CAMEL, AgentVerse, MetaGPT, ChatDev, AutoGen, LLM-Debate, and DyLAN (Li et al., 2023; Chen et al., 2023; Hong et al., 2023; Qian et al., 2024; Wu et al., 2023b; Du et al., 2023a; Liu et al., 2024). Subsequent methods adapt interaction structures through graph induction, multi-agent architecture search, or agent-participation control, such as G-Designer, GPTSwarm, MaAS, Assemble Your Crew, AgentDropout, GTD, and TopoDIM (Zhang et al., 2024b; Zhuge et al., 2024; Zhang et al., 2025b; Li et al., 2025; Wang et al., 2025; Jiang et al., 2025a; Sun et al., 2025). These approaches generally do not jointly learn round-wise communication modes, directed partner selection, and topology-conditioned team-value evaluation. Agent Q-Mix introduces reinforcement-learning-based round-wise communication scheduling with a QMIX mixer (Jiang et al., 2026a), but does not explicitly model agent roles or condition its value factorization on individual directed communication neighborhoods. ACTS addresses these limitations through role-conditioned relational attention and topological attention for directed partner selection and topology-aware value factorization.

2.2 Cooperative Multi-Agent Reinforcement Learning

Cooperative MARL provides the optimization framework for decentralized decision-making under a shared team objective. Centralized training with decentralized execution (CTDE) uses global information during training while restricting execution to local observations (Oliehoek & Amato, 2016). Value-factorization methods, including VDN, QMIX, QPLEX, and Qatten, represent the team value through individual utilities with different mixing structures (Sunehag et al., 2018; Rashid et al., 2018b; Wang et al., 2021; Yang et al., 2020). Differentiable communication methods further study message passing and attention-based aggregation (Sukhbaatar et al., 2016; Foerster et al., 2016). ACTS builds on this foundation by learning discrete communication actions, role-conditioned interaction topologies, and topology-dependent utility weights within a CTDE framework.

3 ACTS: Dual-Attention Communication Topology Scheduling

ACTS is a CTDE framework for adaptive communication-topology scheduling in cooperative LLM agents (Oliehoek & Amato, 2016). It combines value-based mode selection with role-aware relational attention for directed partner selection, and incorporates the resulting topology into centralized value factorization through topological attention. Figure 2 illustrates the overall framework.

3.1 Problem Formulation

We formulate communication-topology scheduling as a cooperative multi-agent decision problem under CTDE. Let $\mathcal{I} = \{1, \dots, N\}$ denote the set of cooperative LLM agents, where agent i is assigned a role ρ_i. At round t, agent i receives a local observation o_i^t and maintains a recurrent state h_i^t encoding its interaction history. The global state s^t is available only during centralized training.

Communication is represented by a directed graph $G^t = (\mathcal{I}, \mathcal{E}^t)$, where $(i, j) \in \mathcal{E}^t$ denotes a communication link from agent i to agent j. We initialize G^1 as the complete directed graph without self-loops and set $\mathcal{I}_{\text{comm}}^1 = \mathcal{I}$. At round t, the agents jointly select $\boldsymbol{a}^t = (a_1^t, \dots, a_N^t)$ with $a_i^t \in \mathcal{A}_{\text{comm}}$. The communication-eligible set for the next round is

$$\mathcal{I}_{\text{comm}}^{t+1} = \left\{ i \in \mathcal{I} \mid a_i^t \notin \{a^{\text{off}}, a^{\text{solo}}\} \right\}. \tag{1}$$

The selected modes and directional attention scores then determine the receiver and sender sets $\mathcal{S}_i^{t+1}$ and $\mathcal{R}_i^{t+1}$, which define the directed topology G^{t+1}.

For subsequent analysis, IG denotes the topology-induced information gain, which measures the information-acquisition effectiveness of the topology scheduler. Let π denote the topology-scheduling policy and $c_{\text{tok}}^t(\tau)$ the token consumption at round t along trajectory τ. The policy is

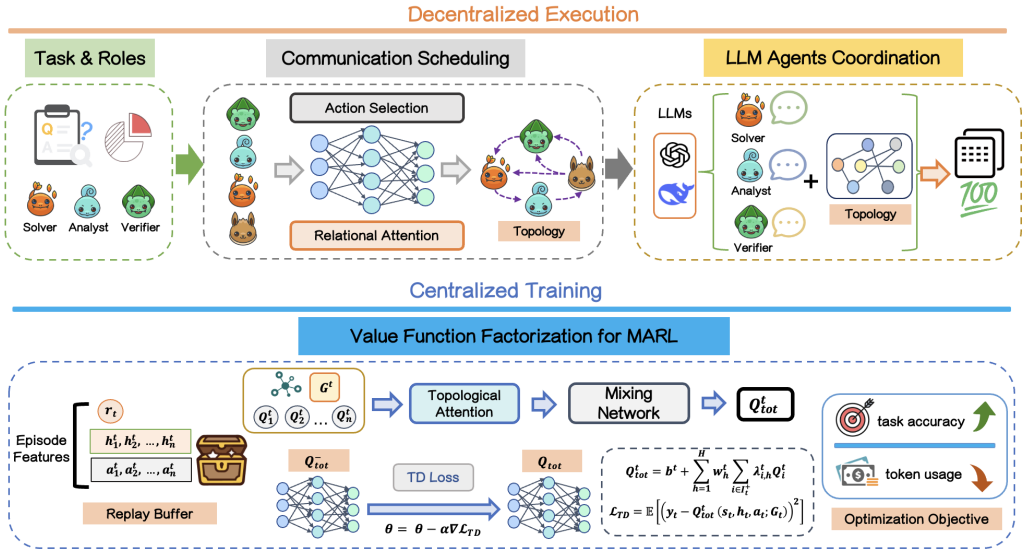


Figure 2: **Overview of ACTS.** At decision round t, the realized topology G^t conditions both the local encoder and the centralized mixer. Relational attention produces local utility features and edge scores. The communication actions then induce G^{t+1} for the next context.

optimized to balance task performance and token efficiency:

$$\pi^\star = \arg\max_{\pi} \mathbb{E}_{\tau\sim\pi}\left[w_{\text{task}} R_{\text{task}}(\tau) - w_{\text{tok}} \min\left(\frac{\sum_{t=1}^{T} c_{\text{tok}}^t(\tau)}{B}, 1\right)\right], \tag{2}$$

where $R_{\text{task}}(\tau)$ denotes the final task score, B is a token-cost normalization scale, and w_{task} and w_{tok} control the trade-off between task performance and token efficiency.

3.2 Relational Attention for Communication Topology Learning

Existing topology schedulers rely on fixed communication patterns and do not explicitly model role-dependent partner relevance (Jiang et al., 2026a). ACTS combines communication-mode selection with relational attention over recurrent representations that encode agent roles and evolving local contexts, thereby constructing adaptive directed neighborhoods with a variable number of partners.

At round t, the GNN aggregates each agent's local observation and role information according to the current topology G^t, while a GRU updates the recurrent state using the previous hidden state:

$$h_i^t = \text{GRU}_{\theta_h}\left(\text{GNN}_{\theta_g}\left(o_i^t, e_\rho(\rho_i), G^t\right), h_i^{t-1}\right), \tag{3}$$

where $e_\rho(\rho_i)$ denotes the embedding of agent i's role. The resulting states are processed by a cross-attention module, which computes the directed relevance between each ordered pair of agents:

$$\alpha_{ij}^t = \text{softmax}_{j\neq i}\left(\frac{\left(W_{Q,\text{rel}} h_i^t\right)^\top \left(W_{K,\text{rel}} h_j^t\right)}{\sqrt{d_{\text{rel}}}}\right). \tag{4}$$

The ordered query–key construction produces directional attention weights, with $\alpha_{ij}^t \neq \alpha_{ji}^t$ in general. The cross-attention layer refines each recurrent state by aggregating value projections from other agents with a residual connection, producing the feature used for local mode selection:

$$\xi_i^t = h_i^t + \sum_{j\in\mathcal{I}\setminus\{i\}} \alpha_{ij}^t W_{V,\text{rel}} h_j^t, \qquad a_i^t = \arg\max_{a\in\mathcal{A}_{\text{comm}}} Q(\xi_i^t, a). \tag{5}$$

The communication mode specifies the direction and scope of interaction, while relational attention selects the corresponding partners. For the eligible candidate set $\mathcal{J}_i^{t+1}$, the receiver and sender sets

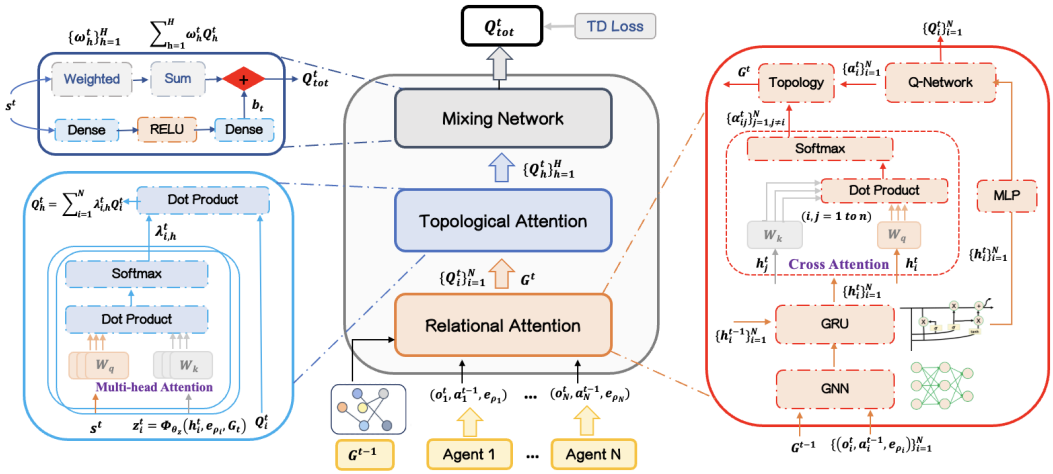


Figure 3: **Architecture of ACTS.** Given G^t, the local branch produces role-aware recurrent representations, individual action values, and directed partner scores. Communication modes and relational scores jointly induce G^{t+1}. During centralized training, the directed neighborhoods in G^t are encoded as z_i^t and used by topological attention to combine individual values into Q_{tot}^t.

are

$$\mathcal{S}_i^{t+1} = \begin{cases} \mathcal{J}_i^{t+1}, & a_i^t = a^{\text{broadcast}}, \\ \{j \in \mathcal{J}_i^{t+1} \mid \alpha_{ij}^t > \bar{\alpha}_i^{\text{out},t}\}, & a_i^t = a^{\text{attn-broadcast}}, \\ \varnothing, & \text{otherwise}, \end{cases}$$
$$\mathcal{R}_i^{t+1} = \begin{cases} \mathcal{J}_i^{t+1}, & a_i^t = a^{\text{receive}}, \\ \{j \in \mathcal{J}_i^{t+1} \mid \alpha_{ji}^t > \bar{\alpha}_i^{\text{in},t}\}, & a_i^t = a^{\text{attn-receive}}, \\ \varnothing, & \text{otherwise}. \end{cases} \tag{6}$$

The resulting directed topology is

$$\mathcal{E}^{t+1} = \left\{(i,j) \in \mathcal{I} \times \mathcal{I} \;\middle|\; i \neq j,\ j \in \mathcal{S}_i^{t+1} \ \vee\ i \in \mathcal{R}_j^{t+1}\right\}, \qquad G^{t+1} = \left(\mathcal{I}, \mathcal{E}^{t+1}\right). \tag{7}$$

By incorporating role embeddings and topology-aware recurrent states, ACTS conditions both action evaluation and partner selection on the current agent context. This decoupled design enables directed, variable-cardinality topologies that adapt across rounds. The complete construction is given in Appendix B.

3.3 Topological Attention for Value Function Factorization

We adopt the standard CTDE value-factorization framework, in which a centralized mixer combines individual utilities into a joint action value, while the IGM property enables decentralized greedy execution (Sunehag et al., 2018; Rashid et al., 2018b). A concise background is provided in Appendix A. Existing state-conditioned mixers, however, derive their mixing coefficients from the global state without explicitly encoding the realized communication topology. For a fixed pre-decision context, the joint value admits the local approximation

$$Q_{\text{tot}}^t \approx b^t + \sum_{i \in \mathcal{I}_{\text{comm}}^t} \beta_i^t Q_i^t, \qquad \beta_i^t = \beta_i\left(s^t, \mathcal{N}_i^{\text{in},t}, \mathcal{N}_i^{\text{out},t}\right), \tag{8}$$

where β_i^t denotes the contribution of agent i's utility under its current directed communication neighborhoods. As shown in Appendix D.1, this coefficient reflects the interaction terms, including higher-order effects, induced by these neighborhoods. ACTS therefore learns topology-dependent mixing coefficients using topological attention.

For each candidate communication action a_i, let $Q_i^t(a_i) = Q_i(\xi_i^t, a_i)$ denote the local action value. ACTS encodes the directed neighborhoods in G^t by

$$\begin{aligned} m_{i,r}^t &= \mathrm{Agg}_r\left(\left\{\phi_r\big(h_j^t \,\|\, e_\rho(\rho_j)\big) \,\middle|\, j \in \mathcal{N}_i^{r,t}\right\}\right), \\ z_i^t &= \Phi_{\theta_z}\left(h_i^t \,\|\, e_\rho(\rho_i) \,\|\, m_{i,\mathrm{in}}^t \,\|\, m_{i,\mathrm{out}}^t\right), \qquad r \in \{\mathrm{in}, \mathrm{out}\}. \end{aligned} \tag{9}$$

The topology representation z_i^t, used only by the centralized mixer, serves as the key in topological attention, with the global state s^t providing the query. For head h, the normalized attention weight is

$$\lambda_{i,h}^t = \frac{\exp\left(\left(W_{Q,\mathrm{top}}^{(h)} s^t\right)^\top \left(W_{K,\mathrm{top}}^{(h)} z_i^t\right)/\sqrt{d_h}\right)}{\sum_{j \in \mathcal{I}_{\mathrm{comm}}^t} \exp\left(\left(W_{Q,\mathrm{top}}^{(h)} s^t\right)^\top \left(W_{K,\mathrm{top}}^{(h)} z_j^t\right)/\sqrt{d_h}\right)}. \tag{10}$$

With nonnegative head weights $w_h^t = |[f_{\theta_w}(s^t)]_h|$ and offset $b^t = V_{\theta_b}(s^t)$, the centralized mixer is

$$Q_{\mathrm{tot}}^t(\boldsymbol{a}) = b^t + \sum_{h=1}^{H} w_h^t \left(\sum_{i \in \mathcal{I}_{\mathrm{comm}}^t} \lambda_{i,h}^t Q_i^t(a_i)\right). \tag{11}$$

Here, $\widehat{\beta}_i^t = \sum_{h=1}^H w_h^t \lambda_{i,h}^t$ is the effective mixing coefficient. Since $w_h^t \geq 0$ and $\lambda_{i,h}^t \geq 0$, $\partial Q_{\mathrm{tot}}^t / \partial Q_i^t(a_i) = \widehat{\beta}_i^t \geq 0$, so the mixer is monotonic in every local action value. Under this fixed-context construction, Appendix D.3 establishes the Individual–Global–Max (IGM) property:

$$\operatorname*{arg\,max}_{\boldsymbol{a} \in \prod_{i \in \mathcal{I}_{\mathrm{comm}}^t} \mathcal{A}_i^t} Q_{\mathrm{tot}}^t(\boldsymbol{a}) = \prod_{i \in \mathcal{I}_{\mathrm{comm}}^t} \operatorname*{arg\,max}_{a_i \in \mathcal{A}_i^t} Q_i^t(a_i). \tag{12}$$

Thus, decentralized greedy maximization is consistent with the centralized objective. Appendix D.2 further shows that $\widehat{\beta}_i^t$ is both the mixer sensitivity to Q_i^t and the scale of the corresponding utility-level TD signal.

3.4 Cooperative MARL Training and Execution

The training and execution procedures are detailed in Appendix E. During decentralized execution, agent i selects $a_i^t = \arg\max_{a \in \mathcal{A}_{\mathrm{comm}}} Q_i(\xi_i^t, a)$, and the joint action $\boldsymbol{a}^t = (a_1^t, \ldots, a_N^t)$ induces the next-round topology G^{t+1}. During centralized training, the mixer evaluates each replayed transition conditioned on its pre-decision state and topology, with the terminal team reward specified by Eq. equation 2.

For each sampled transition, the target joint action is selected by decentralized greedy evaluation:

$$\boldsymbol{a}^{*,t+1} = \left(\operatorname*{arg\,max}_{a \in \mathcal{A}_{\mathrm{comm}}} Q_i^-(\xi_i^{t+1}, a)\right)_{i=1}^N. \tag{13}$$

Using the fixed next-round state and topology, the target value and TD loss are

$$\begin{aligned} y^t &= r^t + \gamma(1 - d^t) Q_{\mathrm{tot}}^-\left(\boldsymbol{a}^{*,t+1}\right), \\ \mathcal{L}_{\mathrm{TD}} &= \mathbb{E}_{\mathcal{D}}\left[\left(y^t - Q_{\mathrm{tot}}^t(\boldsymbol{a}^t)\right)^2\right]. \end{aligned} \tag{14}$$

The GNN, GRU, relational-attention/Q branch, and topological mixer are optimized jointly, while the LLM parameters remain fixed. Mixer quantities are held constant during each candidate-action comparison, and the nondifferentiable topology construction is excluded from backpropagation.

4 Experiments

4.1 Experimental Setup

We evaluate ACTS on nine benchmarks spanning four domains: code generation (LiveCodeBench v6 and HumanEval), multiple-choice reasoning (MMLU-Pro and the first 250 questions of Humanity's Last Exam), multi-hop question answering (HotpotQA), and competition mathematics (AIME

Table 1: Accuracy (%) across nine benchmarks. Bold marks the best result in each column; ACTS is highlighted in light blue. Avg. is the unweighted macro-average over the nine benchmarks.

Method	LiveCode Bench	Human Eval	HLE	Hotpot QA	MMLU-Pro	AIME 2025	AIME 2026	Beyond-AIME	HMMT 2025	Avg.
Base model: GPT-OSS-120B										
Direct Prompting	88.57	89.63	42.40	20.00	70.00	60.00	86.67	16.00	46.67	57.77
Static topology										
LLM-Debate	92.00	88.41	42.80	21.35	71.43	63.33	86.67	18.00	46.67	58.96
AutoGen	95.43	90.24	41.20	32.03	77.14	70.00	80.00	18.00	46.67	61.19
LangGraph	93.14	89.02	42.40	40.00	84.29	76.67	86.67	23.00	50.00	65.02
Agent Framework	93.14	90.24	42.00	32.03	84.29	73.33	76.67	36.00	40.00	63.08
Centralized adaptive topology										
G-Designer	94.29	93.29	42.80	34.73	70.00	73.33	83.33	28.00	43.33	62.57
TopoDIM	93.71	92.68	42.00	30.68	65.71	66.67	83.33	25.00	33.33	59.23
GTD	94.86	93.90	42.40	35.95	71.43	70.00	80.00	30.00	46.67	62.80
MaAS	94.86	93.29	41.60	33.38	68.57	66.67	86.67	27.00	40.00	61.34
Decentralized learned topology										
Agent Q-Mix	95.43	**100.00**	38.40	40.00	**100.00**	80.00	90.00	36.00	66.67	71.83
ACTS	**96.57**	**100.00**	**43.20**	**46.67**	**100.00**	**83.33**	**93.33**	**42.00**	**70.00**	**75.01**
Base model: DeepSeek-V4-Flash										
Direct Prompting	93.14	98.17	33.60	40.00	72.86	50.00	66.67	10.00	46.67	56.79
Static topology										
LLM-Debate	93.14	96.95	34.00	40.00	61.43	53.33	66.67	12.00	36.67	54.91
AutoGen	**96.00**	98.17	32.80	42.70	71.43	60.00	70.00	15.00	46.67	59.20
LangGraph	92.57	97.56	33.60	46.62	78.57	60.00	70.00	26.00	46.67	61.29
Agent Framework	89.71	98.17	33.20	44.05	75.71	66.67	76.67	20.00	43.33	60.83
Centralized adaptive topology										
G-Designer	93.71	97.56	33.60	46.62	88.57	63.33	73.33	22.00	43.33	62.45
TopoDIM	92.57	97.56	32.80	44.05	88.57	60.00	70.00	18.00	46.67	61.14
GTD	93.71	97.56	34.00	47.97	90.00	63.33	73.33	24.00	43.33	63.03
MaAS	94.29	98.17	32.80	46.62	88.57	63.33	73.33	20.00	43.33	62.27
Decentralized learned topology										
Agent Q-Mix	93.14	**100.00**	32.80	44.00	92.86	60.00	73.33	33.00	40.00	63.24
ACTS	**96.00**	**100.00**	**34.40**	**50.67**	**97.14**	**70.00**	**86.67**	**40.00**	**50.00**	**69.43**

2025/2026, HMMT 2025, and Beyond-AIME) (Jain et al., 2024; Chen et al., 2021; Wang et al., 2024; The Humanity's Last Exam Collaboration, 2025; Yang et al., 2018; Mathematical Association of America, 2025; MathArena, 2025; ByteDance Seed, 2025). We use GPT-OSS-120B and DeepSeek-V4-Flash as frozen backbone LLMs; each task instantiates three role-conditioned agents and a final answer-aggregation node. Baselines include direct prompting, fixed or framework-based coordination (LLM-Debate, AutoGen, LangGraph, and Agent Framework), task-adaptive topology scheduling (TopoDIM), and value-based MARL scheduling (Agent Q-Mix) (Du et al., 2023a; Wu et al., 2023b; LangChain AI, 2024; Sun et al., 2025; Jiang et al., 2026b). All methods use the same prompts, team size, communication-round budget, and evaluation protocol. Results are averaged over three independent random seeds. Detailed benchmark, baseline, and training configurations are provided in Appendix F.1.

4.2 Main Results

Accuracy gains from selective communication. Table 1 reports accuracy across all nine benchmarks. ACTS achieves the highest average accuracy on both backbone models, 75.01% on GPT-OSS-120B and 69.43% on DeepSeek-V4-Flash, exceeding Agent Q-Mix by 3.18 and 6.19 points in each case, and attains the best or tied-best result on every benchmark. This advantage is concentrated on tasks where agents hold complementary information, because relational attention selects communication partners by semantic relevance rather than by fixed topology or content-agnostic index rules.

Pareto-optimal efficiency across IG regimes. Figure 4 plots accuracy against token consumption on HotpotQA (IG $>$ 0) and HLE (IG $\approx$ 0); full counts are provided in Table 3 (Appendix F.5). Static topologies broadcast every round regardless of whether communication is useful; centralized

adaptive topologies adjust the structure only at generation time, not during execution. Agent Q-Mix learns agent deactivation and primitive-based routing, but discrete communication primitives provide limited selectivity. ACTS achieves Pareto optimality across all four settings: it matches or exceeds Agent Q-Mix at lower token cost and dominates static and centralized families on both accuracy and cost. Only single-agent prompting uses fewer tokens, at an accuracy cost of approximately 27 percentage points on HotpotQA with GPT-OSS-120B. On HLE, where information gain is near zero, ACTS and single-agent prompting yield comparable performance. By learning whether communication is warranted and routing through attention affinities, ACTS spends tokens only where expected information gain justifies the expenditure.

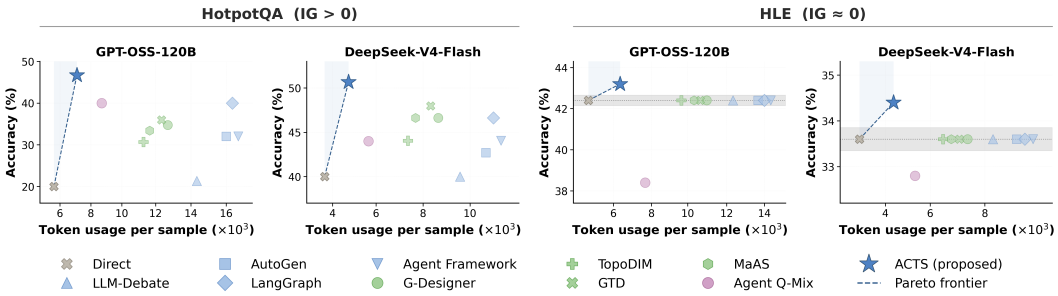


Figure 4: Accuracy vs. token usage for all methods on two representative tasks (HotpotQA with IG $>$ 0 and HLE with IG $\approx$ 0) and two backbone models. The dashed line traces the Pareto frontier of methods not dominated on both accuracy and cost. ACTS lies on the frontier in all four panels: no other multi-agent method reaches its accuracy at lower or equal cost, and only single-agent prompting is cheaper, at a substantial accuracy loss.

4.3 Experimental Analysis

4.3.1 Information Gain Analysis

An operational criterion for Information Gain. Information gain depends on model capability, task characteristics, and communication efficiency. Keeping topology fixed, Figure 5 compares the three topology families by mean accuracy gain over single-agent prompting. We define IG ≈ 0 when the observed uplift falls within sampling noise, which corresponds to approximately two questions under the $x = 2/n$ criterion applied over n samples. The model-task pair is classified as IG ≈ 0 if all three families fall below this threshold, and as IG > 0 otherwise; formally, for model m and task t, we write $\mathrm{IG}(m,t) \in \{\approx 0, > 0\}$.

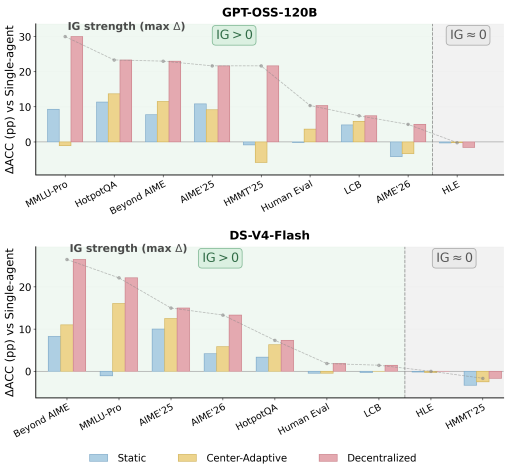


Figure 5: Mean ΔACC of the three topology families relative to single-agent prompting on two backbone LLMs. Benchmarks to the left of the dashed line are IG > 0; those to the right are IG ≈ 0 (within $\pm 2/n$ sampling noise).

Communication utility is gated by Information Gain. Information gain is positive when agents supply complementary evidence that no single perspective can provide. This occurs on HotpotQA, where agents independently gather evidence on distinct entities that subsequently enables cross-comparison, and on contest mathematics, where competing algebraic, geometric, and computational approaches yield differentiable solution paths. Information gain is null when a single perspective suffices. On HLE the underlying model's knowledge ceiling renders inter-agent communication redundant. Under the same task, HMMT yields IG ≈ 0 with DeepSeek-V4-Flash because limited model capacity produces unreliable and undifferentiated outputs, leaving the attention module without discriminative signal.

ACTS is designed to operate in both regimes: it maximizes communication utility where IG > 0 through attention-driven partner selection, and it degrades to solo inference where IG ≈ 0.

4.3.2 Communication Pattern Analysis

The information gains of ACTS arise from two complementary mechanisms: attention-guided selective communication and dynamic deactivation. We analyze their effects on a high-IG task (HotpotQA) and a low-IG task (HLE).

Attention-guided selective communication. At each round, the agent network produces asymmetric directional scores α_{ij}^t, indicating the relevance of agent j to agent i. The *attn-broadcast* action retains the directed edge $i \to j$ when the outgoing score α_{ij}^t exceeds its mean over the eligible candidates, whereas *attn-receive* retains $j \to i$ when the corresponding incoming score α_{ji}^t exceeds the same threshold. These actions therefore construct sparse, round-specific neighborhoods rather than using a fixed graph. Since the attention module is trained end-to-end with the accuracy–token objective, the policy can suppress redundant interactions, yielding the token savings reported in Table 3 (Appendix F.5).

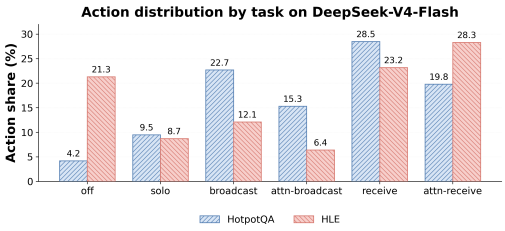


Figure 6: Action distribution by task on DeepSeek-V4-Flash. On HotpotQA, the policy sustains agent activity and broadcast communication; on HLE, it deactivates a substantial fraction of rounds and relies on pull-based retrieval.

Dynamic deactivation. The *off* action prunes communication further by deactivating an agent, which then makes no LLM call and emits no communication edges. When a majority of agents become inactive, the episode terminates early, avoiding additional discussion rounds that are unlikely to change the answer. Thus, tasks that require only limited collaboration can approach single-agent cost, whereas agents remain active when broader collaboration is useful.

Action profiles across the two information-gain regimes. Figure 6 contrasts the action distributions learned on HotpotQA and HLE under DeepSeek-V4-Flash. On HotpotQA the policy keeps most agents active: the off share remains at 4.2%, while broadcast reaches 22.7%. On HLE the profile inverts: off rises to 21.3%, and broadcast and attn-broadcast decline to 12.1% and 6.4%, with attn-receive accounting for 28.3% of the actions. A trajectory-level view of these two regimes is provided in Appendix F.4.

4.3.3 Ablation Studies

Table 2 evaluates the three components that distinguish ACTS from Agent Q-Mix. We consider: (a) replacing the topological attention mixer with QMIX; (b) replacing learned relational routing with fixed role-based partners; and (c) removing role embeddings from relational attention. All other components and training settings remain unchanged. Relational attention produces the directional scores α_{ij}^t used by the attention-guided communication modes, whereas the topological attention mixer determines the topology-dependent weights in joint-value factorization.

Table 2: Ablation results on the two backbone models. $\overline{\text{ACC}}$ and $\overline{\text{Token}}$ denote the macro-average accuracy (%) and macro-average per-sample token usage ($\times 10^3$) across the nine benchmarks. The full ACTS model is highlighted.

	GPT-OSS-120B		DeepSeek-V4-Flash	
Variant	$\overline{\text{ACC}}$	$\overline{\text{Token}}$	$\overline{\text{ACC}}$	$\overline{\text{Token}}$
ACTS	**75.01**	**5.97**	**69.43**	**4.03**
(a) Topological mixer replaced by QMIX	71.83	7.41	63.24	4.96
(b) Learned routing replaced by fixed role-based partners	72.50	8.10	66.10	6.20
(c) Role embeddings removed from relational attention	73.40	6.80	67.10	4.60

Topology-aware value factorization. Replacing the topological attention mixer with QMIX reduces macro-average accuracy from 75.01% to 71.83% on GPT-OSS-120B and from 69.43% to 63.24% on DeepSeek-V4-Flash. Token usage also increases from 5.97 to 7.41 and from 4.03 to 4.96, respectively. This degradation indicates that conditioning utility weights on directed neighborhood representations improves both team-value estimation and communication efficiency.

Adaptive partner selection. Replacing learned relational routing with fixed role-based partners reduces accuracy to 72.50% and 66.10%, while increasing token usage to 8.10 and 6.20 on the two backbones. Fixed assignments cannot adapt partner selection to the current task and may continue communication when the assigned partner provides limited additional information. This variant therefore produces the largest increase in token usage.

Role-conditioned routing. Removing role embeddings from relational attention yields accuracies of 73.40% and 67.10%, with token usage increasing to 6.80 and 4.60. Because the attention mechanisms remain intact, the degradation is smaller than that caused by replacing the topology-aware mixer. Nevertheless, the consistent decrease in accuracy and increase in token usage show that role information improves the selectivity of directed partner selection.

5 Conclusion

6 AI use statement

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, "we checked LLM-generated research ideas for potential plagiarism through a manual literature survey", "LLM-generated code was verified and tested for correctness by 2 authors", etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

See the ICLR 2027 AI Policy for Authors for more details. This statement should not be more than 1 page.

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A Value Function Factorization in Cooperative Multi-Agent Reinforcement Learning

A.1 Centralized Training with Decentralized Execution

Cooperative multi-agent reinforcement learning (MARL) considers a set of agents that jointly maximize a shared return. Let s^t denote the global state, τ_i^t the local action-observation history of agent i, and $\boldsymbol{u}^t = (u_1^t, \ldots, u_N^t)$ the joint action. The objective is

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=1}^{T} \gamma^{t-1} r^t \right], \tag{15}$$

where r^t is the shared reward and $\gamma \in [0, 1)$ is the discount factor.

In partially observable environments, each agent makes decisions from its local action-observation history. Independent value-based learning is challenging because the joint action space grows exponentially with the number of agents, while the effective environment of each agent changes as the other agents update their policies. Centralized training with decentralized execution (CTDE) addresses these issues by allowing the training procedure to access centralized information, while restricting execution to local histories.

Let $Q_i(\tau_i^t, u_i)$ denote the individual action-value function of agent i. Under CTDE, the centralized joint action-value function can be represented as

$$Q_{\text{tot}}\left(s^t, \boldsymbol{\tau}^t, \boldsymbol{u}^t\right) = F_\psi\left(s^t, \left\{Q_i(\tau_i^t, u_i^t)\right\}_{i=1}^N\right), \tag{16}$$

where F_ψ is a centralized factorization function. This formulation allows centralized training to model inter-agent dependencies while retaining decentralized action selection at execution time.

A.2 Value Function Factorization

Value Decomposition Networks (VDN) use the additive factorization

$$Q_{\text{tot}}\left(\boldsymbol{\tau}, \boldsymbol{u}\right) = \sum_{i=1}^{N} Q_i(\tau_i, u_i), \tag{17}$$

which supports decentralized execution but cannot represent general non-additive interactions among agents (Sunehag et al., 2018).

QMIX introduces a nonlinear factorization function (Rashid et al., 2018b):

$$Q_{\text{tot}} = F_\psi\left(Q_1, \ldots, Q_N; s\right), \tag{18}$$

subject to the monotonicity condition

$$\frac{\partial Q_{\text{tot}}}{\partial Q_i} \geq 0, \qquad i = 1, \ldots, N. \tag{19}$$

Thus, increasing an individual utility cannot decrease the joint value when the remaining utilities and the centralized state are fixed. QMIX typically enforces this condition by generating nonnegative factorization weights from the global state through hypernetworks.

Subsequent methods improve the expressiveness of value function factorization in different ways. QTRAN introduces transformation and consistency constraints to represent a broader class of joint value functions (Son et al., 2019). QPLEX uses a duplex dueling architecture to construct a more expressive value–advantage factorization while preserving decentralized consistency (Wang et al., 2021). Attention- based factorization methods assign context-dependent weights to individual utilities, allowing their contributions to vary with the centralized context (Yang et al., 2020).

A.3 Monotonic Value Mixing and the IGM Property

A factorized joint action-value function satisfies the Individual–Global–Max (IGM) property if its globally greedy action is consistent with the individual greedy actions:

$$\arg\max_{\boldsymbol{u}} Q_{\mathrm{tot}}\left(s, \boldsymbol{\tau}, \boldsymbol{u}\right) = \left(\arg\max_{u_1} Q_1(\tau_1, u_1), \ldots, \arg\max_{u_N} Q_N(\tau_N, u_N)\right). \tag{20}$$

With consistent tie-breaking, this property means that the globally optimal joint action can be recovered by independently maximizing the individual action-value functions.

Componentwise monotonicity is a sufficient condition for IGM. For fixed s and $\boldsymbol{\tau}$, consider two joint actions that differ only in agent i's action. If one action produces a larger value of $Q_i(\tau_i, u_i)$, the monotonicity condition ensures that it cannot produce a smaller value of Q_{tot}. Applying this argument coordinate-wise to all agents shows that independent greedy action selection yields a globally optimal joint action.

A.4 Temporal-Difference Optimization

Value-based MARL methods are commonly trained with a centralized temporal-difference objective. For a transition $(s^t, \boldsymbol{\tau}^t, \boldsymbol{u}^t, r^t, s^{t+1}, d^t)$, where d^t indicates episode termination, a target network forms

$$y^t = r^t + \gamma(1 - d^t) Q^-_{\mathrm{tot}}\left(s^{t+1}, \boldsymbol{\tau}^{t+1}, \boldsymbol{u}^{*,t+1}\right). \tag{21}$$

When the factorization satisfies IGM, the target joint action can be selected independently:

$$u_i^{*,t+1} = \arg\max_{u_i} Q_i^-\left(\tau_i^{t+1}, u_i\right), \qquad i = 1, \ldots, N. \tag{22}$$

The online parameters are optimized by minimizing

$$\mathcal{L}_{\mathrm{TD}} = \mathbb{E}_{\mathcal{D}}\left[\left(y^t - Q_{\mathrm{tot}}\left(s^t, \boldsymbol{\tau}^t, \boldsymbol{u}^t\right)\right)^2\right]. \tag{23}$$

Experience replay and delayed target networks are commonly used to stabilize optimization under deep function approximation.

B Construction of the Directed Communication Topology

This section gives the complete construction of $G^{t+1} = \Psi(\boldsymbol{a}^t, \boldsymbol{\alpha}^t)$. After communication-mode selection, define the communication-eligible agents and candidate partners as

$$\mathcal{I}_{t+1}^{\mathrm{comm}} = \left\{ i \in \mathcal{I} \;\middle|\; a_i^t \notin \left\{a^{\mathrm{off}}, a^{\mathrm{solo}}\right\} \right\}, \qquad \mathcal{J}_i^{t+1} = \mathcal{I}_{t+1}^{\mathrm{comm}} \setminus \{i\}. \tag{24}$$

The index $t+1$ indicates that these sets determine the topology produced by the decisions at round t.

For $i \in \mathcal{I}_{t+1}^{\mathrm{comm}}$ with $\mathcal{J}_i^{t+1} \neq \varnothing$, define the outgoing and incoming mean scores as

$$\bar{\alpha}_i^{\mathrm{out},t} = \frac{1}{|\mathcal{J}_i^{t+1}|} \sum_{j \in \mathcal{J}_i^{t+1}} \alpha_{ij}^t, \qquad \bar{\alpha}_i^{\mathrm{in},t} = \frac{1}{|\mathcal{J}_i^{t+1}|} \sum_{j \in \mathcal{J}_i^{t+1}} \alpha_{ji}^t. \tag{25}$$

The requested receiver and sender sets are

$$\mathcal{S}_i^{t+1} = \begin{cases} \mathcal{J}_i^{t+1}, & a_i^t = a^{\text{broadcast}}, \\ \left\{ j \in \mathcal{J}_i^{t+1} \;\middle|\; \alpha_{ij}^t > \bar{\alpha}_i^{\text{out},t} \right\}, & a_i^t = a^{\text{attn-broadcast}}, \\ \varnothing, & \text{otherwise}, \end{cases}$$
$$\mathcal{R}_i^{t+1} = \begin{cases} \mathcal{J}_i^{t+1}, & a_i^t = a^{\text{receive}}, \\ \left\{ j \in \mathcal{J}_i^{t+1} \;\middle|\; \alpha_{ji}^t > \bar{\alpha}_i^{\text{in},t} \right\}, & a_i^t = a^{\text{attn-receive}}, \\ \varnothing, & \text{otherwise}. \end{cases} \tag{26}$$

When $\mathcal{J}_i^{t+1} = \varnothing$, both request sets are defined as empty.

The directed edge set is then

$$\mathcal{E}^{t+1} = \left\{ (i,j) \in \mathcal{I}_{t+1}^{\text{comm}} \times \mathcal{I}_{t+1}^{\text{comm}} \;\middle|\; i \neq j,\ j \in \mathcal{S}_i^{t+1} \vee i \in \mathcal{R}_j^{t+1} \right\}, \tag{27}$$

and

$$G^{t+1} = \left(\mathcal{I}, \mathcal{E}^{t+1} \right). \tag{28}$$

Thus, $a^{\text{broadcast}}$ and a^{receive} request all eligible interactions in the corresponding direction, whereas $a^{\text{attn-broadcast}}$ and $a^{\text{attn-receive}}$ retain only interactions whose scores exceed the corresponding mean. The strict mean threshold allows the number of selected partners to adapt to the score distribution without introducing a fixed top-k hyperparameter.

C Theoretical Analysis of Topological Value Factorization

At a non-terminal round t, let $\mathcal{I}_{\text{comm}}^t$ denote the nonempty set of agents eligible to make communication decisions. This set is fixed before the current actions are selected. The corresponding pre-decision context is

$$\mathcal{C}^t = \left(s^t, G^t, \mathcal{I}_{\text{comm}}^t, \{\xi_i^t, z_i^t\}_{i \in \mathcal{I}_{\text{comm}}^t} \right), \tag{29}$$

where ξ_i^t is the input to the local action-value function and z_i^t encodes agent i's directed neighborhoods in G^t for centralized value function factorization. Let $\boldsymbol{a}^t = (a_i^t)_{i \in \mathcal{I}}$ denote the full joint action, with $a_i^t = a^{\text{off}}$ for agents outside $\mathcal{I}_{\text{comm}}^t$. The actions selected at round t determine the communication-eligible set for the next round:

$$\mathcal{I}_{\text{comm}}^{t+1} = \left\{ i \in \mathcal{I} : a_i^t \notin \{a^{\text{off}}, a^{\text{solo}}\} \right\}. \tag{30}$$

Given G^t, $\boldsymbol{a}^t$, and the directional scores $\boldsymbol{\alpha}^t$, the sender and receiver sets, and hence the directed topology G^{t+1}, are constructed according to the procedure in Appendix B.

C.1 Local Expansion and Directed-Neighborhood Dependence

Fix the pre-decision context $\mathcal{C}^t$ and let

$$\boldsymbol{q} = (q_i)_{i \in \mathcal{I}_{\text{comm}}^t}$$

denote a vector of individual utility values. Suppose that the reference joint-value function is locally represented by a twice continuously differentiable map

$$F_{\mathcal{C}^t} : \mathcal{U} \subseteq \mathbb{R}^{|\mathcal{I}_{\text{comm}}^t|} \to \mathbb{R}.$$

This smoothness assumption is imposed in utility space and does not require the discrete communication actions to be differentiable.

For $\boldsymbol{u} \in \mathcal{U}$, define the local sensitivity

$$\beta_i^{\text{ref},t}(\boldsymbol{u}) = \partial_i F_{\mathcal{C}^t}(\boldsymbol{u}).$$

For a reference point $\bar{\boldsymbol{q}} \in \mathcal{U}$, let

$$\beta_i^{\text{ref},t} = \beta_i^{\text{ref},t}(\bar{\boldsymbol{q}}), \qquad c^{\text{ref},t} = F_{\mathcal{C}^t}(\bar{\boldsymbol{q}}) - \sum_{i \in \mathcal{I}_{\text{comm}}^t} \beta_i^{\text{ref},t} \bar{q}_i. \tag{31}$$

Proposition C.1 (Local weighted expansion)**.** *If the line segment between $\bar{\boldsymbol{q}}$ and $\boldsymbol{q}$ lies in $\mathcal{U}$, then*

$$F_{\mathcal{C}^t}(\boldsymbol{q}) = c^{\mathrm{ref},t} + \sum_{i \in \mathcal{I}^t_{\mathrm{comm}}} \beta_i^{\mathrm{ref},t} q_i + R_2^t. \tag{32}$$

Moreover, if

$$\left\| \nabla^2 F_{\mathcal{C}^t}(\boldsymbol{u}) \right\|_{2\to 2} \le L$$

along this segment, then

$$\left| R_2^t \right| \le \frac{L}{2} \left\| \boldsymbol{q} - \bar{\boldsymbol{q}} \right\|_2^2. \tag{33}$$

Proof. Let

$$\boldsymbol{\Delta} = \boldsymbol{q} - \bar{\boldsymbol{q}}.$$

The integral form of Taylor's theorem gives

$$\begin{aligned} F_{\mathcal{C}^t}(\boldsymbol{q}) = F_{\mathcal{C}^t}(\bar{\boldsymbol{q}}) + \nabla F_{\mathcal{C}^t}(\bar{\boldsymbol{q}})^\top \boldsymbol{\Delta} \\ + \int_0^1 (1-u)\, \boldsymbol{\Delta}^\top \nabla^2 F_{\mathcal{C}^t}\left(\bar{\boldsymbol{q}} + u\boldsymbol{\Delta}\right) \boldsymbol{\Delta}\, du. \end{aligned} \tag{34}$$

Since $\Delta_i = q_i - \bar{q}_i$ and $\beta_i^{\mathrm{ref},t} = \partial_i F_{\mathcal{C}^t}(\bar{\boldsymbol{q}})$, the first-order terms satisfy

$$F_{\mathcal{C}^t}(\bar{\boldsymbol{q}}) + \nabla F_{\mathcal{C}^t}(\bar{\boldsymbol{q}})^\top \boldsymbol{\Delta} = c^{\mathrm{ref},t} + \sum_{i \in \mathcal{I}^t_{\mathrm{comm}}} \beta_i^{\mathrm{ref},t} q_i.$$

This proves Eq. equation 53. Furthermore,

$$\begin{aligned} \left| R_2^t \right| &\le \int_0^1 (1-u) \left\| \nabla^2 F_{\mathcal{C}^t}\left(\bar{\boldsymbol{q}} + u\boldsymbol{\Delta}\right) \right\|_{2\to 2} \left\| \boldsymbol{\Delta} \right\|_2^2 \, du \\ &\le L \left\| \boldsymbol{\Delta} \right\|_2^2 \int_0^1 (1-u)\, du = \frac{L}{2} \left\| \boldsymbol{q} - \bar{\boldsymbol{q}} \right\|_2^2. \end{aligned}$$

□

The coefficient $\beta_i^{\mathrm{ref},t}$ is the local sensitivity of the reference joint value to agent i's utility. Its dependence on other utilities is governed by mixed derivatives. For any $\boldsymbol{\delta}$ such that the corresponding line segment lies in $\mathcal{U}$,

$$\begin{aligned} &\beta_i^{\mathrm{ref},t}\left(\bar{\boldsymbol{q}} + \boldsymbol{\delta}\right) - \beta_i^{\mathrm{ref},t}\left(\bar{\boldsymbol{q}}\right) \\ &\quad = \sum_{j \in \mathcal{I}^t_{\mathrm{comm}}} \delta_j \int_0^1 \partial_{ij} F_{\mathcal{C}^t}\left(\bar{\boldsymbol{q}} + u\boldsymbol{\delta}\right) du. \end{aligned} \tag{35}$$

Proof. Define

$$g(u) = \partial_i F_{\mathcal{C}^t}\left(\bar{\boldsymbol{q}} + u\boldsymbol{\delta}\right).$$

By the chain rule,

$$g'(u) = \sum_{j \in \mathcal{I}^t_{\mathrm{comm}}} \delta_j \partial_{ij} F_{\mathcal{C}^t}\left(\bar{\boldsymbol{q}} + u\boldsymbol{\delta}\right).$$

The fundamental theorem of calculus therefore gives

$$g(1) - g(0) = \int_0^1 g'(u)\, du,$$

which proves Eq. equation 56. □

Thus, mixed derivatives describe how the utility of agent j changes the local contribution of agent i.

To make the dependence on the communication topology explicit, define

$$\mathcal{E}^t_{\mathrm{comm}} = \mathcal{E}^t \cap \left(\mathcal{I}^t_{\mathrm{comm}} \times \mathcal{I}^t_{\mathrm{comm}}\right),$$

and the directed neighborhoods

$$\mathcal{N}_i^{\mathrm{out},t} = \{j : (i,j) \in \mathcal{E}_{\mathrm{comm}}^t\}, \qquad \mathcal{N}_i^{\mathrm{in},t} = \{j : (j,i) \in \mathcal{E}_{\mathrm{comm}}^t\}.$$

More generally, let $\mathcal{S}(G^t)$ denote a collection of graph-supported scopes and write $\boldsymbol{q}_S = (q_j)_{j\in S}$. Suppose that

$$F_{\mathcal{C}^t}(\boldsymbol{q}) = a_{\mathcal{C}^t} + \sum_{S\in\mathcal{S}(G^t)} f_S(\boldsymbol{q}_S; G^t). \tag{36}$$

Then

$$\beta_i^{\mathrm{ref},t}(\boldsymbol{q}) = \sum_{\substack{S\in\mathcal{S}(G^t)\\ i\in S}} \partial_i f_S(\boldsymbol{q}_S; G^t). \tag{37}$$

Proof. Differentiating Eq. equation 36 with respect to q_i eliminates every factor whose scope does not contain agent i. The remaining factors give Eq. equation 37. □

Hence, every graph-induced interaction factor containing agent i can affect its local coefficient, including higher-order neighborhood interactions.

For the pairwise edge-factor special case,

$$F_{\mathcal{C}^t}(\boldsymbol{q}) = a_{\mathcal{C}^t} + \sum_{i\in\mathcal{I}_{\mathrm{comm}}^t} f_i(q_i) + \sum_{(i,j)\in\mathcal{E}_{\mathrm{comm}}^t} f_{ij}(q_i, q_j). \tag{38}$$

Differentiating with respect to q_i gives

$$\begin{aligned}\beta_i^{\mathrm{ref},t}(\boldsymbol{q}) = f_i'(q_i) &+ \sum_{j\in\mathcal{N}_i^{\mathrm{out},t}} \partial_1 f_{ij}(q_i, q_j) \\ &+ \sum_{j\in\mathcal{N}_i^{\mathrm{in},t}} \partial_2 f_{ji}(q_j, q_i).\end{aligned} \tag{39}$$

Proof. The unary term $f_i(q_i)$ contributes $f_i'(q_i)$. For every outgoing edge (i,j), q_i is the first argument of f_{ij} and contributes $\partial_1 f_{ij}(q_i, q_j)$. For every incoming edge (j,i), q_i is the second argument of f_{ji} and contributes $\partial_2 f_{ji}(q_j, q_i)$. All other edge factors are independent of q_i and therefore contribute zero. □

This pairwise case makes explicit why both directed incoming and outgoing neighborhoods can affect an agent's contribution. The reference coefficient $\beta_i^{\mathrm{ref},t}$ need not be nonnegative. ACTS instead learns the nonnegative topology-dependent coefficient $\widehat{\beta}_i^t$ through TD optimization.

C.2 Topology-Aware TD Credit Assignment

For each agent $i \in \mathcal{I}_{\mathrm{comm}}^t$, define

$$Q_i^t(a_i) = Q_i(\xi_i^t, a_i).$$

For head h, the topological-attention logit and normalized weight are

$$\eta_{i,h}^t = \frac{\left(W_{Q,\mathrm{top}}^{(h)} s^t\right)^\top \left(W_{K,\mathrm{top}}^{(h)} z_i^t\right)}{\sqrt{d_h}}, \qquad \lambda_{i,h}^t = \frac{\exp(\eta_{i,h}^t)}{\sum_{j\in\mathcal{I}_{\mathrm{comm}}^t}\exp(\eta_{j,h}^t)}. \tag{40}$$

Let $w_h^t = |[f_{\theta_w}(s^t)]_h| \geq 0$ and $b^t = V_{\theta_b}(s^t)$. The centralized mixer is

$$\begin{aligned}Q_{\mathrm{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) &= b^t + \sum_{h=1}^H w_h^t \left(\sum_{i\in\mathcal{I}_{\mathrm{comm}}^t} \lambda_{i,h}^t Q_i^t(a_i)\right) \\ &= b^t + \sum_{i\in\mathcal{I}_{\mathrm{comm}}^t} \widehat{\beta}_i^t Q_i^t(a_i),\end{aligned} \tag{41}$$

where

$$\widehat{\beta}_i^t = \sum_{h=1}^{H} w_h^t \lambda_{i,h}^t. \tag{42}$$

For a sampled transition, let y^t denote the detached TD target and define

$$\delta^t = Q_{\mathrm{tot}}^t(\mathcal{C}^t, \boldsymbol{a}^t) - y^t, \qquad \ell_{\mathrm{TD}}^t = \frac{1}{2}(\delta^t)^2.$$

The following are partial derivatives with respect to the mixer inputs, conditioned on the current pre-decision context.

Proposition C.2 (Topology-aware TD gradients)**.** *Holding the current context fixed,*

$$\frac{\partial \ell_{\mathrm{TD}}^t}{\partial Q_i^t(a_i^t)} = \delta^t \widehat{\beta}_i^t, \tag{43}$$

and, with the local utilities held fixed,

$$\frac{\partial \ell_{\mathrm{TD}}^t}{\partial \eta_{i,h}^t} = \delta^t w_h^t \lambda_{i,h}^t \left(Q_i^t(a_i^t) - Q_h^t(\boldsymbol{a}^t) \right), \tag{44}$$

where

$$Q_h^t(\boldsymbol{a}^t) = \sum_{j \in \mathcal{I}_{\mathrm{comm}}^t} \lambda_{j,h}^t Q_j^t(a_j^t).$$

Proof. The mixer gives

$$\frac{\partial Q_{\mathrm{tot}}^t}{\partial Q_i^t(a_i^t)} = \widehat{\beta}_i^t.$$

Since y^t is detached, Eq. equation 64 follows from the chain rule. For the attention derivative, the softmax Jacobian is

$$\frac{\partial \lambda_{j,h}^t}{\partial \eta_{i,h}^t} = \lambda_{j,h}^t \left(\mathbb{1}\{i = j\} - \lambda_{i,h}^t \right).$$

Therefore,

$$\frac{\partial Q_h^t}{\partial \eta_{i,h}^t} = \lambda_{i,h}^t \left(Q_i^t(a_i^t) - Q_h^t(\boldsymbol{a}^t) \right),$$

which gives Eq. equation 65. □

Thus, $\widehat{\beta}_i^t$ is both the mixer sensitivity to agent i's selected utility and the scale of its direct utility-level TD signal. Because $\widehat{\beta}_i^t$ depends on z_i^t, and z_i^t encodes the directed neighborhoods in G^t, this credit assignment is topology-aware.

The discrete topology-construction map is not differentiated through. The selected action determines G^{t+1} and the next context, but does not change the mixer coefficients while candidate actions at round t are being compared.

C.3 Monotonicity and Individual–Global–Max Consistency

Let every active agent have a finite, nonempty action space $\mathcal{A}_i^t$, and define

$$\mathcal{A}^t = \prod_{i \in \mathcal{I}_{\mathrm{comm}}^t} \mathcal{A}_i^t. \tag{45}$$

The actions a^{off} and a^{solo} remain valid candidates in the current comparison. Their effects are reflected only in $\mathcal{I}_{\mathrm{comm}}^{t+1}$ and G^{t+1} after the joint action has been selected.

Because $w_h^t \geq 0$ and $\lambda_{i,h}^t \geq 0$,

$$\frac{\partial Q_{\mathrm{tot}}^t}{\partial Q_i^t(a_i)} = \widehat{\beta}_i^t \geq 0. \tag{46}$$

Thus, the mixer is monotonic in every local action value.

Define

$$\mathcal{M}_i^t = \arg\max_{a_i \in \mathcal{A}_i^t} Q_i^t(a_i), \qquad \mathcal{P}^t = \{i : \widehat{\beta}_i^t > 0\}, \qquad \mathcal{Z}^t = \{i : \widehat{\beta}_i^t = 0\}.$$

Theorem C.3 (IGM consistency)**.** *For the fixed context $\mathcal{C}^t$,*

$$\arg\max_{\boldsymbol{a} \in \mathcal{A}^t} Q_{\text{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) = \left(\prod_{i \in \mathcal{P}^t} \mathcal{M}_i^t \right) \times \left(\prod_{i \in \mathcal{Z}^t} \mathcal{A}_i^t \right). \tag{47}$$

If all attention logits are finite and

$$\sum_{h=1}^{H} w_h^t > 0, \tag{48}$$

then $\widehat{\beta}_i^t > 0$ for every $i \in \mathcal{I}_{\text{comm}}^t$, and

$$\arg\max_{\boldsymbol{a} \in \mathcal{A}^t} Q_{\text{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) = \prod_{i \in \mathcal{I}_{\text{comm}}^t} \arg\max_{a_i \in \mathcal{A}_i^t} Q_i^t(a_i). \tag{49}$$

Proof. Let

$$m_i^t = \max_{a_i \in \mathcal{A}_i^t} Q_i^t(a_i).$$

For any joint action,

$$Q_{\text{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) \le b^t + \sum_{i \in \mathcal{I}_{\text{comm}}^t} \widehat{\beta}_i^t m_i^t.$$

Equality holds exactly when every agent with $\widehat{\beta}_i^t > 0$ selects an action in $\mathcal{M}_i^t$; actions of agents in $\mathcal{Z}^t$ do not affect the joint value. This proves Eq. equation 71.

Finite logits imply $\lambda_{i,h}^t > 0$ for every agent and head. If $\sum_h w_h^t > 0$, at least one head has positive weight, and therefore

$$\widehat{\beta}_i^t = \sum_{h=1}^{H} w_h^t \lambda_{i,h}^t > 0$$

for every $i \in \mathcal{I}_{\text{comm}}^t$. Hence $\mathcal{Z}^t = \varnothing$, which yields Eq. equation 73. □

If all head weights are zero, then $Q_{\text{tot}}^t = b^t$ for every candidate joint action. Every joint action, including every independently greedy joint action, is therefore globally optimal, although the exact set-valued identity in Eq. equation 73 need not hold.

The result depends on the fixed-context construction: all candidate actions at round t share the same pre-decision topology and mixer coefficients, whereas the selected action determines only the topology for the next round.

D Theoretical Analysis of Topological Value Factorization

At round t, let

$$\mathcal{C}^t = \left(s^t, G^t, \{\xi_i^t, z_i^t\}_{i \in \mathcal{D}^t}, \mathcal{D}^t\right) \tag{50}$$

denote the pre-decision context, where $\mathcal{D}^t$ is the fixed, nonempty set of agents making communication decisions. Here, ξ_i^t is used by the individual action-value function, whereas z_i^t encodes the directed neighborhoods in G^t for centralized value factorization. The selected joint action subsequently induces

$$G^{t+1} = \Psi(\mathcal{C}^t, \boldsymbol{a}^t). \tag{51}$$

D.1 Local Expansion and Interaction-Dependent Coefficients

Fix $\mathcal{C}^t$ and let $\boldsymbol{q} = (q_i)_{i\in\mathcal{D}^t}$ denote the individual action values. Suppose that a reference joint action-value function is represented locally by a smooth map

$$F_{\mathcal{C}^t} : \mathcal{U} \to \mathbb{R}, \qquad \mathcal{U} \subseteq \mathbb{R}^{|\mathcal{D}^t|}.$$

This assumption is made in utility space and therefore does not require the discrete communication actions to be differentiable.

Proposition D.1 (Local weighted expansion)**.** *Let $F_{\mathcal{C}^t} \in C^2(\mathcal{U})$, and let the line segment between $\bar{\boldsymbol{q}}$ and $\boldsymbol{q}$ lie in $\mathcal{U}$. Define*

$$\beta_i^{\mathrm{ref}} = \partial_i F_{\mathcal{C}^t}(\bar{\boldsymbol{q}}), \qquad c^{\mathrm{ref}} = F_{\mathcal{C}^t}(\bar{\boldsymbol{q}}) - \sum_{i\in\mathcal{D}^t} \beta_i^{\mathrm{ref}} \bar{q}_i. \tag{52}$$

Then

$$F_{\mathcal{C}^t}(\boldsymbol{q}) = c^{\mathrm{ref}} + \sum_{i\in\mathcal{D}^t} \beta_i^{\mathrm{ref}} q_i + R_2. \tag{53}$$

If $\|\nabla^2 F_{\mathcal{C}^t}(\boldsymbol{u})\|_{2\to 2} \le L$ along the segment, then

$$|R_2| \le \frac{L}{2}\|\boldsymbol{q} - \bar{\boldsymbol{q}}\|_2^2. \tag{54}$$

Proof. Let $\boldsymbol{\Delta} = \boldsymbol{q} - \bar{\boldsymbol{q}}$. Taylor's theorem gives

$$\begin{aligned} F_{\mathcal{C}^t}(\boldsymbol{q}) &= F_{\mathcal{C}^t}(\bar{\boldsymbol{q}}) + \nabla F_{\mathcal{C}^t}(\bar{\boldsymbol{q}})^\top \boldsymbol{\Delta} \\ &\quad + \int_0^1 (1-u)\, \boldsymbol{\Delta}^\top \nabla^2 F_{\mathcal{C}^t}(\bar{\boldsymbol{q}} + u\boldsymbol{\Delta})\boldsymbol{\Delta}\, du. \end{aligned} \tag{55}$$

Substituting $\Delta_i = q_i - \bar{q}_i$ yields Eq. equation 53. The Hessian bound and $\int_0^1 (1-u)\, du = 1/2$ give Eq. equation 54. □

The coefficient β_i^{ref} is the local sensitivity of the reference joint value to agent i's utility. Its variation across operating points reflects inter-agent interactions. In particular, for a displacement $\boldsymbol{\delta}$,

$$\begin{aligned} &\beta_i^{\mathrm{ref}}(\bar{\boldsymbol{q}} + \boldsymbol{\delta}) - \beta_i^{\mathrm{ref}}(\bar{\boldsymbol{q}}) \\ &\quad = \sum_{j\in\mathcal{D}^t} \delta_j \int_0^1 \partial_{ij} F_{\mathcal{C}^t}(\bar{\boldsymbol{q}} + u\boldsymbol{\delta})\, du. \end{aligned} \tag{56}$$

Thus, for $j \neq i$, the mixed derivative $\partial_{ij} F_{\mathcal{C}^t}$ describes how agent j changes the local coefficient of agent i. This motivates learning agent-specific mixing coefficients (Yang et al., 2020).

A direct connection to communication neighborhoods follows under a graph-structured reference value. Let

$$\mathcal{E}_{\mathcal{D}}^t = \mathcal{E}^t \cap (\mathcal{D}^t \times \mathcal{D}^t)$$

be the directed, self-loop-free edge set induced on $\mathcal{D}^t$.

Corollary D.2 (Directed-neighborhood dependence)**.** *Suppose*

$$F_{\mathcal{C}^t}(\boldsymbol{q}) = a_{\mathcal{C}^t} + \sum_{i\in\mathcal{D}^t} f_i(q_i) + \sum_{(i,j)\in\mathcal{E}_{\mathcal{D}}^t} f_{ij}(q_i, q_j). \tag{57}$$

Then

$$\begin{aligned} \beta_i^{\mathrm{ref}}(\boldsymbol{q}) &= f_i'(q_i) + \sum_{j\in\mathcal{N}_i^{\mathrm{out},t}} \partial_1 f_{ij}(q_i, q_j) \\ &\quad + \sum_{j\in\mathcal{N}_i^{\mathrm{in},t}} \partial_2 f_{ji}(q_j, q_i). \end{aligned} \tag{58}$$

Proof. Differentiate Eq. equation 57 with respect to q_i. Only the unary factor of agent i and the edge factors incident to i have nonzero derivatives. □

The corollary establishes locality with respect to utility coordinates under the graph-structured assumption. It motivates using incoming and outgoing neighborhood representations to estimate the mixing coefficients. ACTS does not compute β_i^{ref} analytically; it learns a nonnegative surrogate from TD targets.

D.2 Topology-Aware TD Credit Assignment

Define

$$q_i^t(a_i) = Q_i(\xi_i^t, a_i), \qquad \eta_{i,h}^t = \frac{(W_q^h s^t)^\top (W_k^h z_i^t)}{\sqrt{d_h}}, \tag{59}$$

and

$$\lambda_{i,h}^t = \frac{\exp(\eta_{i,h}^t)}{\sum_{j\in\mathcal{D}^t}\exp(\eta_{j,h}^t)}. \tag{60}$$

The head-level and joint action values are

$$\begin{aligned} Q_h^t(\boldsymbol{a}) &= \sum_{i\in\mathcal{D}^t} \lambda_{i,h}^t q_i^t(a_i), \\ Q_{\mathrm{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) &= b^t + \sum_{h=1}^H w_h^t Q_h^t(\boldsymbol{a}) \\ &= b^t + \sum_{i\in\mathcal{D}^t} \widehat{\beta}_i^t q_i^t(a_i), \end{aligned} \tag{61}$$

where

$$\widehat{\beta}_i^t = \sum_{h=1}^H w_h^t \lambda_{i,h}^t, \qquad w_h^t \geq 0. \tag{62}$$

For a sampled transition, let y^t be the detached TD target and set

$$\delta^t = Q_{\mathrm{tot}}^t - y^t, \qquad \ell_{\mathrm{TD}}^t = \frac{1}{2}(\delta^t)^2. \tag{63}$$

Proposition D.3 (Topology-aware TD gradients)**.** *Conditional on the current context $\mathcal{C}^t$,*

$$\frac{\partial \ell_{\mathrm{TD}}^t}{\partial q_i^t(a_i^t)} = \delta^t \widehat{\beta}_i^t, \tag{64}$$

and

$$\frac{\partial \ell_{\mathrm{TD}}^t}{\partial \eta_{i,h}^t} = \delta^t w_h^t \lambda_{i,h}^t \left(q_i^t(a_i^t) - Q_h^t(\boldsymbol{a}^t)\right). \tag{65}$$

Proof. Equation equation 61 gives

$$\frac{\partial Q_{\mathrm{tot}}^t}{\partial q_i^t} = \widehat{\beta}_i^t.$$

Since the TD target is detached, Eq. equation 64 follows by the chain rule. Moreover,

$$\frac{\partial \lambda_{j,h}^t}{\partial \eta_{i,h}^t} = \lambda_{j,h}^t \left(\mathbb{I}[i=j] - \lambda_{i,h}^t\right), \tag{66}$$

which implies

$$\frac{\partial Q_h^t}{\partial \eta_{i,h}^t} = \lambda_{i,h}^t \left(q_i^t(a_i^t) - Q_h^t(\boldsymbol{a}^t)\right). \tag{67}$$

Combining this identity with $\partial Q_{\mathrm{tot}}^t / \partial Q_h^t = w_h^t$ proves Eq. equation 65. □

Equation equation 64 shows that $\widehat{\beta}_i^t$ is both the mixer Jacobian with respect to agent i's selected utility and the scale of its direct TD signal. Because $\widehat{\beta}_i^t$ depends on z_i^t and hence on G^t, this credit allocation is topology-aware.

These equations are computational-graph partial derivatives conditional on the current graph. They do not differentiate the discrete topology-construction map Ψ. The topology G^{t+1} generated by $\boldsymbol{a}^t$ affects the next context and the detached TD target, rather than receiving a gradient through the discrete graph transition.

D.3 Monotonicity and IGM Consistency

Let each agent have a finite, nonempty action space $\mathcal{A}_i^t$, and assume that

$$\mathcal{A}^t = \prod_{i \in \mathcal{D}^t} \mathcal{A}_i^t. \tag{68}$$

The current context and mixing coefficients are fixed when actions in $\mathcal{A}^t$ are compared. In particular, a^{off} is an ordinary candidate action whose effect is reflected in G^{t+1}, rather than in the current attention mask.

From Eq. equation 62,

$$\frac{\partial Q_{\text{tot}}^t}{\partial q_i^t} = \widehat{\beta}_i^t \geq 0, \tag{69}$$

so the mixer is monotonic in every individual utility.

Define

$$\mathcal{M}_i^t = \arg\max_{a_i \in \mathcal{A}_i^t} q_i^t(a_i), \qquad \mathcal{P}^t = \{i : \widehat{\beta}_i^t > 0\}, \qquad \mathcal{Z}^t = \{i : \widehat{\beta}_i^t = 0\}. \tag{70}$$

Theorem D.4 (IGM consistency)**.** *For the fixed context $\mathcal{C}^t$,*

$$\arg\max_{\boldsymbol{a} \in \mathcal{A}^t} Q_{\text{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) = \left(\prod_{i \in \mathcal{P}^t} \mathcal{M}_i^t \right) \times \left(\prod_{i \in \mathcal{Z}^t} \mathcal{A}_i^t \right). \tag{71}$$

Consequently, every independently greedy joint action is a joint maximizer. If

$$W^t := \sum_{h=1}^{H} w_h^t > 0 \tag{72}$$

and the attention logits are finite, then $\widehat{\beta}_i^t > 0$ for every $i \in \mathcal{D}^t$, and the set-valued IGM identity holds:

$$\arg\max_{\boldsymbol{a} \in \mathcal{A}^t} Q_{\text{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) = \prod_{i \in \mathcal{D}^t} \arg\max_{a_i \in \mathcal{A}_i^t} Q_i(\xi_i^t, a_i). \tag{73}$$

Proof. Let $m_i^t = \max_{a_i \in \mathcal{A}_i^t} q_i^t(a_i)$. For every joint action $\boldsymbol{a}$,

$$Q_{\text{tot}}^t(\mathcal{C}^t, \boldsymbol{a}) \leq b^t + \sum_{i \in \mathcal{D}^t} \widehat{\beta}_i^t m_i^t. \tag{74}$$

Equality holds exactly when $q_i^t(a_i) = m_i^t$ for every $i \in \mathcal{P}^t$; actions of agents in $\mathcal{Z}^t$ do not affect the joint value. This proves Eq. equation 71.

Finite logits imply $\lambda_{i,h}^t > 0$. If $W^t > 0$, some head $h^\star$ satisfies $w_{h^\star}^t > 0$, and hence

$$\widehat{\beta}_i^t \geq w_{h^\star}^t \lambda_{i,h^\star}^t > 0 \tag{75}$$

for every agent. Therefore $\mathcal{Z}^t = \varnothing$, and Eq. equation 71 reduces to Eq. equation 73. □

If all head weights are zero, the joint action value equals b^t for every candidate action. Independently greedy actions remain joint maximizers, although the two maximizing sets need not be identical. The condition $W^t > 0$ can be enforced, if needed, through a strictly positive head-weight parameterization.

The IGM result relies on the temporal ordering in Eqs. equation 50–equation 51: all candidate actions at round t share the same current context and mixing coefficients, while the selected action determines the next topology. This is the condition under which monotonicity yields decentralized greedy consistency.

E ACTS Training and Decentralized Execution

We use the GNN–GRU encoder and relational-attention/Q branch defined in Sec. 3.2. The topology encoder and topological mixer follow Sec. 3.3. The operator BuildTopology denotes the construction in Appendix B; in particular, broadcast and receive modes retain all eligible partners, whereas attention-guided modes retain candidates whose directional scores exceed the corresponding candidate-set mean.

The mixer quantities used for a transition are computed from the pre-decision state and topology and remain fixed while candidate actions are evaluated. The topology induced by a candidate target action is not constructed during the same target maximization. The target branch is detached from the online update. Since hard partner-set construction is nondifferentiable, relational attention receives TD supervision through its soft feature branch, whereas topological attention is trained through the centralized mixer. Replay stores raw inputs and topologies rather than cached hidden representations.

F Experimental Details

F.1 Training and Evaluation Details

Benchmarks and backbone models. We evaluate `ACTS` on nine benchmarks across four domains. The coding benchmarks are LiveCodeBench v6 and HumanEval (Jain et al., 2024; Chen et al., 2021). The reasoning benchmarks are MMLU-Pro and the first 250 questions of Humanity's Last Exam (Wang et al., 2024; The Humanity's Last Exam Collaboration, 2025). HotpotQA is used for multi-hop question answering (Yang et al., 2018). The mathematical benchmarks are AIME 2025, AIME 2026, HMMT 2025, and Beyond-AIME (Mathematical Association of America, 2025; MathArena, 2025; ByteDance Seed, 2025). Experiments use GPT-OSS-120B and DeepSeek-V4-Flash as frozen backbone LLMs. Each task is instantiated with three role-conditioned agents followed by a final answer-aggregation node.

Baselines and controlled comparison. We compare `ACTS` with direct prompting, LLM-Debate (Du et al., 2023a), AutoGen (Wu et al., 2023b), LangGraph (LangChain AI, 2024), the evaluated Agent Framework implementation, TopoDIM (Sun et al., 2025), and Agent Q-Mix (Jiang et al., 2026b). All methods use the same team size ($N = 3$), task prompts, communication-round budget, and answer-evaluation procedure.

Training protocol and evaluation. The ACTS scheduler is trained for 50 episodes using 15 training examples per domain. The communication budget is three rounds for mathematical tasks and two rounds for coding and reasoning tasks. All trainable modules are optimized with Adam using learning rate 5×10^{-4} and discount factor $\gamma = 0.99$. The replay buffer has capacity $5{,}000$, the mini-batch size is 8, and ϵ-greedy exploration is annealed linearly from 1.0 to 0.05 over 50 episodes. The learned parameters are fixed during evaluation. Reported results are averaged over three independent runs with different random seeds.

F.2 States and Observations

`ACTS` follows a centralised-training, decentralised-execution scheme. A shared per-agent value network receives a local observation o_i for each agent i, while a mixing network integrates the agent-specific values into a joint estimate Q_{tot} conditioned on a global state s. We describe the construction of each.

Local observation. At every round each agent i receives a 54-dimensional observation $o_i \in \mathbb{R}^{54}$ for the $N = 3$ setting, assembled as the concatenation of five blocks: (i) a 32-dimensional task

Algorithm 1 ACTS Centralized Training

Require: Training tasks $\mathcal{D}_{\text{train}}$, episodes M, rounds T, discount factor γ, target-update interval K
Ensure: Online parameters Θ and target parameters Θ^{-}

Initialize Θ, set $\Theta^{-} \leftarrow \Theta$, initialize replay buffer $\mathcal{D}$, and set $\epsilon \leftarrow \epsilon_0$
for $e = 1, \dots, M$ **do**
  Sample $x \sim \mathcal{D}_{\text{train}}$ and initialize G^1, $\mathcal{I}^1_{\text{comm}} = \mathcal{I}$, and h_i^0 for all agents
  Initialize an empty episode trajectory
  for $t = 1, \dots, T$ **do**
    Obtain local inputs $\{o_i^t\}$ under G^t
    for $i \in \mathcal{I}^t_{\text{comm}}$ **do**
      Compute $h_i^t \leftarrow \mathrm{GRU}(\mathrm{GNN}(o_i^t, e_\rho(\rho_i), G^t), h_i^{t-1})$
    end for
    Compute directed weights $\boldsymbol{\alpha}^t$, features $\{\xi_i^t\}$, and local values $Q_i^t(a) = Q_i(\xi_i^t, a)$; select a_i^t by ϵ-greedy exploration
    Form $\boldsymbol{a}^t = (a_1^t, \dots, a_N^t)$ and set $\mathcal{I}^{t+1}_{\text{comm}} \leftarrow \{i : a_i^t \notin \{a^{\text{off}}, a^{\text{solo}}\}\}$
    Construct $G^{t+1} \leftarrow \textsc{BuildTopology}(G^t, \boldsymbol{a}^t, \boldsymbol{\alpha}^t)$
    Execute $(\boldsymbol{a}^t, G^{t+1})$ and observe the next inputs, state, token usage, and termination flag d^t
    Store the raw transition $(s^t, \{o_i^t\}, G^t, \mathcal{I}^t_{\text{comm}}, \boldsymbol{a}^t, G^{t+1}, d^t)$
    if $d^t = 1$ **then**
      break
    end if
  end for
  Compute $R \leftarrow w_{\text{task}} R_{\text{task}} - w_{\text{tok}} \min(C_{\text{tok}}/B, 1)$, assign $r^t = 0$ to non-terminal transitions and $r^t = R$ to the terminal transition
  Add the completed episode to $\mathcal{D}$
  if an optimization update is scheduled **then**
    Sample a minibatch and reconstruct the online and target representations from the stored raw inputs
    for each sampled transition **do**
      Evaluate $Q^t_{\text{tot}}(\boldsymbol{a}^t)$ using $(s^t, G^t, \mathcal{I}^t_{\text{comm}})$
      For $i \in \mathcal{I}^{t+1}_{\text{comm}}$, select $a_i^{*,t+1} \leftarrow \arg\max_{a \in \mathcal{A}_{\text{comm}}} Q_i^{-}(\xi_i^{t+1}, a)$
      Evaluate the target mixer once using $(s^{t+1}, G^{t+1}, \mathcal{I}^{t+1}_{\text{comm}})$ and $\boldsymbol{a}^{*,t+1}$
      Set $y^t \leftarrow r^t + \gamma(1 - d^t) Q^{-}_{\text{tot}}(\boldsymbol{a}^{*,t+1})$
    end for
    Minimize $\mathcal{L}_{\text{TD}} \leftarrow \mathbb{E}_{\mathcal{D}}[(y^t - Q^t_{\text{tot}}(\boldsymbol{a}^t))^2]$
    Jointly update the trainable MARL modules and keep the LLM parameters fixed
  end if
  if the number of updates is divisible by K **then**
    Set $\Theta^{-} \leftarrow \Theta$
  end if
  Anneal ϵ
end for

Algorithm 2 ACTS Decentralized Execution

Require: Task query x, role assignments $\{\rho_i\}_{i=1}^N$, trained parameters, and horizon T
Ensure: Final team answer Y

Initialize G^1 as the complete directed graph without self-loops, set $\mathcal{I}^1_{\text{comm}} = \mathcal{I}$, and initialize h_i^0
for $t = 1, \dots, T$ **do**
  Obtain local observations $\{o_i^t\}$ from x, local histories, and messages available under G^t
  for $i \in \mathcal{I}^t_{\text{comm}}$ **do**
    Compute $h_i^t \leftarrow \mathrm{GRU}(\mathrm{GNN}(o_i^t, e_\rho(\rho_i), G^t), h_i^{t-1})$
    Obtain the directed attention weights $\{\alpha_{ij}^t\}$ and local feature ξ_i^t
    Select $a_i^t \leftarrow \arg\max_{a \in \mathcal{A}_{\text{comm}}} Q_i(\xi_i^t, a)$
  end for
  Construct $\mathcal{I}^{t+1}_{\text{comm}}$ and set $G^{t+1} \leftarrow \textsc{BuildTopology}(G^t, \boldsymbol{a}^t, \boldsymbol{\alpha}^t)$
  Execute the induced directed communication schedule and update the agents' local histories
  if the task terminates **then**
    break
  end if
end for
Aggregate the final agent outputs and return Y

feature embedding shared across all agents; (ii) a 16-dimensional one-hot encoding of the agent

index, which is the sole agent-identity input to the value network; (iii) a binary *has-output* flag indicating whether agent i has already produced an answer; (iv) its incident edge count in the current topology; and (v) a token-usage ratio, the cumulative token expenditure of the team so far clipped to $[0, 1]$. The behavioural *role* labels (Independent solver, Team solver, Questioner, Refiner, Verifier, Debater) are injected into the LLM system prompt and therefore act on the trajectory, not on the state vector.

Global state. The mixing network is conditioned on a global state s that concentrates the information needed to combine the per-agent values. It is the concatenation of three parts: the flattened observation tensor of all N agents, so the centralised state contains every local observation; three graph-structure statistics, namely the total edge count, the density, and the agent count; and the flattened $N \times N$ cross-attention matrix a_{edge} produced by the attention module, which encodes pairwise affinities between agents. For $N = 3$ the global state is therefore 174-dimensional.

F.3 Action Space

At each round every active agent selects one of six discrete actions, grouped into three roles. Three actions, *off*, *solo*, and *broadcast*, operate outside the attention matrix: off deactivates the agent entirely for the round, emitting no edges and issuing no LLM call, and once a majority of agents are inactive the episode terminates early; solo produces an independent answer and emits no edges; broadcast sends the agent's current output to every other agent. Two actions build sparse directed edges from the learned attention matrix a_{edge}: *attn-broadcast* opens an edge $i \to j$ when the attention that agent j pays to i exceeds the row mean, and *attn-receive* requests information only from the neighbours that agent i itself attends to. The remaining action, *receive*, performs a generic information fetch. Figure 7 reports the learned distribution of these six actions across the nine benchmarks.

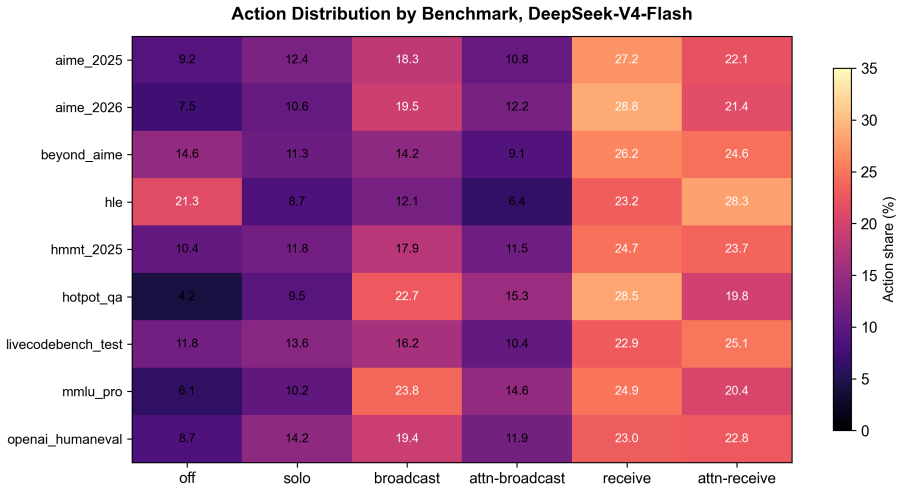


Figure 7: Distribution of the six actions across benchmarks under DeepSeek-V4-Flash. Each cell gives the proportion of rounds in which the corresponding action is selected. The learned policy favours broadcast and attn-broadcast on benchmarks where the agents contribute complementary information, and shifts toward off and attn-receive where a single perspective suffices.

F.4 Case Study

Figure 8 illustrates the same machinery at the trajectory level on one HotpotQA question and one HLE question. On HotpotQA the agents remain active throughout: several agents take the broadcast action, sending their analysis of the question to the team while one agent retains the solo action as an independent reference, after which the broadcasting agents switch to attn-receive, so that the initially connected topology contracts into a sparse pull while the informative edges are preserved. On HLE one agent settles the answer under the solo action, a single attn-broadcast check is routed only to the agent that requires it, and the remaining agents are successively deactivated through the off action, with the majority-inactive condition terminating the episode early. The two trajectories correspond to the two mechanisms that the aggregate action shares in Figure 6 average over: selective routing

Table 3: Average per-sample token usage ($\times 10^3$) on HotpotQA, a high-information-gain (IG > 0) task, and HLE, a low-information-gain (IG ≈ 0) task, for both backbone models. ACTS (highlighted) consumes the fewest tokens among all multi-agent methods in every column; only single-agent prompting uses fewer, at a large accuracy cost.

	HotpotQA		HLE	
Method	GPT-OSS-120B	DeepSeek-V4-Flash	GPT-OSS-120B	DeepSeek-V4-Flash
Direct Prompting	5.8	3.6	4.5	2.8
LLM-Debate	14.1	9.5	12.3	8.2
AutoGen	16.0	10.8	13.8	9.3
LangGraph	16.4	11.0	14.0	9.5
Agent Framework	16.7	11.3	14.2	9.7
G-Designer	12.5	8.4	10.8	7.2
TopoDIM	11.4	7.6	9.8	6.6
GTD	12.1	8.1	10.5	7.0
MaAS	11.8	7.9	10.2	6.8
Agent Q-Mix	8.7	5.8	7.9	5.3
ACTS	7.1	4.8	6.4	4.3

when the agents contribute complementary information, and deactivation when a single perspective suffices.

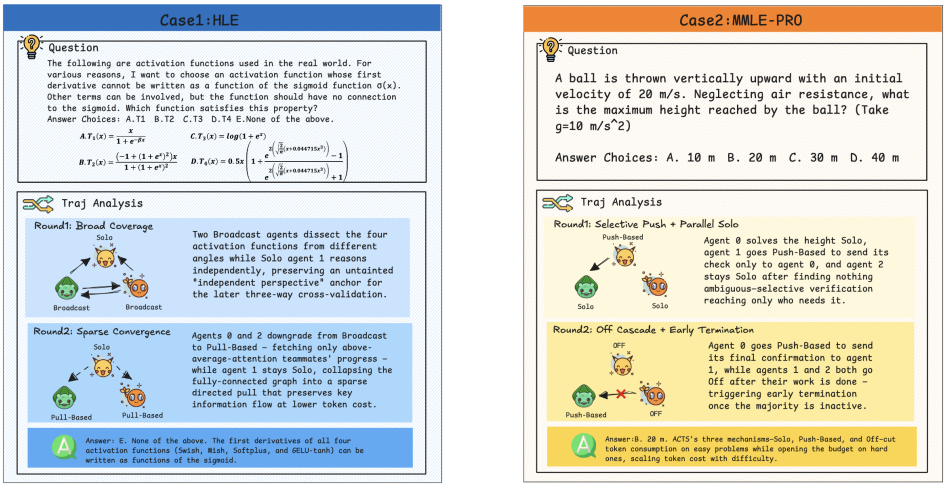


Figure 8: Trajectory-level case study on one HotpotQA question and one HLE question. On HotpotQA the connected topology contracts into a sparse pull while one agent retains the solo action as an independent reference. On HLE one agent settles the answer under the solo action, a single attn-broadcast check is routed to the agent that requires it, and the remaining agents are deactivated through the off action, triggering early termination under the majority-inactive condition.

F.5 Per-Sample Token Usage

Table 3 lists the average per-sample token consumption of each method on HotpotQA, a high-information-gain (IG > 0) task, and HLE, a low-information-gain (IG ≈ 0) task, for both backbone models. These are the numbers plotted against accuracy in Figure 4. Single-agent prompting uses the fewest tokens in every column, but at the cost of substantially lower accuracy; ACTS consumes the fewest tokens among all multi-agent methods in every column, while the static topologies are the most expensive because they broadcast the full running transcript to every agent each round.

F.6 Agent Role Assignments and Prompts

Each agent in the multi-agent system is assigned one of six predefined roles. Roles are allocated in round-robin fashion by agent index modulo 6. Since our experiments use $n = 3$ agents, only the first three roles are instantiated in any given run:

R1: Independent Solver — solves the problem autonomously without relying on peer input.

R2: Team Solver — approaches the problem collaboratively, integrating suggestions from teammates.

R3: Questioner — probes assumptions and raises critical questions about the problem.

The remaining three roles (Refiner, Verifier, Debater) are defined in the role pool for future extensibility to larger agent teams, or for experiments where $n > 3$ (e.g., $n = 6$). The round-robin allocation modulo 6 naturally activates them when the team size increases, without any change to the prompt construction logic.

F.6.1 Agent System Prompt

Every agent receives a system prompt that instantiates its role within an n-agent team and instructs it on the expected output format. The general template is:

`You are {ROLE} on a {n}-agent team. {SUFFIX}`

where `{ROLE}` is one of the six role labels listed above and `{SUFFIX}` is a question-type-specific instruction detailed below.

F.6.2 Question-Type Suffixes

The suffix appended to each agent's system prompt depends on the `answer_type` of the question. Five answer types are supported:

`numerical`. *"Solve the problem step by step, then state the final numerical answer. End with a line 'Final Answer: <number>'."*

`multiple_choice`. *"This is a single-choice question; there is exactly one correct option. The options may be labeled with any letter from A up to Z (not just A–E), so read all options carefully before choosing. Briefly analyze why each candidate is right or wrong, then commit to exactly one option. End with a line 'Final Answer: <letter>' (e.g., 'Final Answer: G')."*

`text`. *"Answer concisely. If a number, give the number. If a name, give the name. End with a line 'Final Answer: <answer>'."*

`open_answer`. *"Answer concisely in a few words. End with a line 'Final Answer: <answer>'."*

`code`. *"Write the complete, runnable solution code. Output ONLY the code—no markdown fences, no explanation."*

F.6.3 User Message

The user message delivered to each agent is constructed by concatenating the following components in order:

1. The question text.
2. Answer options (only for `multiple_choice` questions), formatted as `A. ...`, `B. ...`, etc.
3. Reference passages, when provided by the dataset (e.g., supporting documents for multi-hop QA).
4. Previously generated answers from inbound neighbors in the communication topology, listed as `Agent {i}: {answer}`.

F.6.4 FINAL DECISION MAKER

After all agents have executed across all rounds, a **Final Decision Maker** reviews the collected answers and produces the single final output. Its system prompt is:

> `You are the Final Decision Maker of a {`n`}-agent team. Read the task and all agents' answers carefully, then give the correct single answer.`

The corresponding user message includes the original question, options (if applicable), reference context, and the full list of agent answers, followed by the instruction `Give the final answer.`

F.6.5 NO-ROLE BASELINE

For ablation experiments, we also evaluate a variant where all agents share an identical, role-agnostic system prompt:

> `You are a general team member on a {`n`}-agent team. {SUFFIX}`

This removes the role specialization signal while keeping all other aspects of the prompt (question-type suffix, user message structure, and communication topology) unchanged.